Algebraic Geometry – Background Exercise Sheet

Modules over rings and projective linear algebra

Optional preparation and reference. The module exercises are meant to make explicit which vector-space habits remain valid and which ones fail; the projective exercises provide basic coordinate practice.

Difficulty. ⋆ routine; ⋆⋆ synthesis; ⋆⋆⋆ challenge or extension.
Use. This sheet is optional and independent of the numbered weekly sheets. Cerulean exercises form the suggested route; green exercises are alternatives.
Hints. Read them progressively and stop as soon as one is enough to restart the problem.

Part I — Modules over a ring

Exercise 1. First examples of modules ⋆ **CORE**

Let A be a commutative ring.

(i) Check directly that A^n is an A-module, that every ideal $I \subset A$ is an A-submodule of A, and that A/I is naturally an A-module.

(ii) Explain why a $\mathbb{Z}$-module is the same thing as an abelian group.

(iii) Decide which of the following subsets are submodules:

$$2\mathbb{Z} \subset \mathbb{Z}, \qquad \mathbb{Z}_{\geq 0} \subset \mathbb{Z}, \qquad \Delta = \{(a,a) \mid a \in A\} \subset A^2.$$

Hint 1 (conceptual). A submodule must first be an additive subgroup; only after that does one check stability under multiplication by scalars.

Exercise 2. Bases need not exist ⋆ **CORE**

Regard $\mathbb{Z}/2\mathbb{Z}$ as a $\mathbb{Z}$-module.

(i) Show that every non-zero element of $\mathbb{Z}/2\mathbb{Z}$ satisfies a non-trivial linear relation over $\mathbb{Z}$. Deduce that this non-zero module has no basis.

(ii) Show that $2\mathbb{Z}$ is a free $\mathbb{Z}$-module of rank one.

(iii) Prove that $2\mathbb{Z}$ has no complementary submodule in $\mathbb{Z}$; in other words, there is no submodule $N \subset \mathbb{Z}$ such that

$$\mathbb{Z} = 2\mathbb{Z} \oplus N.$$

(iv) Deduce that the quotient map $\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ has no $\mathbb{Z}$-linear section.

Hint 1 (conceptual). Compare with three familiar facts for vector spaces: every vector space has a basis, every subspace has a complement, and every surjective linear map splits. Each of them can fail for modules.

Hint 2 (technical). For the last part, suppose that $s : \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}$ is a section and apply s to the relation $2\overline{1} = 0$.

Exercise 3. Torsion elements ⋆ **CORE**

Let A be a domain and M an A-module. An element $m \in M$ is called *torsion* if $am = 0$ for some non-zero $a \in A$.

(i) Prove that the torsion elements form a submodule $M_{\text{tor}} \subset M$.

(ii) Determine the torsion submodule of

$$M = \mathbb{Z}/12\mathbb{Z} \oplus \mathbb{Z}.$$

(iii) Compute the orders of $(\overline{4}, 0)$ and $(\overline{3}, 0)$, and decide whether $(\overline{2}, 5)$ is torsion.

(iv) Give a non-zero torsion element in the $\Bbbk[x]$-module $\Bbbk[x]/(x^2)$.

Hint 1 (technical). If $am = 0$ and $bn = 0$, then the non-zero element ab kills $m + n$ because A is a domain.

Exercise 4. Generators and relations ⋆⋆ **CORE**

Let $A = \Bbbk[x, y]$ and $I = (x, y) \subset A$. Consider

$$\varphi : A^2 \longrightarrow I, \qquad (a, b) \longmapsto ax + by.$$

(i) Show that φ is surjective.

(ii) Check that $(-y, x) \in \ker(\varphi)$.

(iii) Prove that every relation between the generators x, y is a multiple of this one:

$$\ker(\varphi) = A(-y, x).$$

(iv) Write the resulting exact sequence. What is the point of contrast with two linearly independent vectors spanning a two-dimensional vector space?

Hint 1 (conceptual). A generating family of a module need not be linearly independent. It is often useful to record a module by generators together with all relations among them.

Hint 2 (technical). If $ax + by = 0$, then x divides by. Since x and y are coprime in the unique factorisation domain $\Bbbk[x, y]$, conclude that x divides b.

Part II — Projective linear algebra

Exercise 5. Projective spans ⋆ **CORE**

For non-zero vectors $v_0, \ldots, v_r \in \Bbbk^{n+1}$, define the projective span of the points $[v_i] \in \mathbb{P}^n_{\Bbbk}$ to be

$$\mathbb{P}\big(\langle v_0, \ldots, v_r \rangle\big).$$

(i) Explain why this does not depend on the chosen representatives v_i.

(ii) In $\mathbb{P}^3_{\Bbbk}$, let

$$p = [1:0:0:0], \quad q = [0:1:0:0], \quad r = [0:0:1:0].$$

Find equations for the line $\langle p, q \rangle$ and for the plane $\langle p, q, r \rangle$.

(iii) Does the point $s = [1:1:1:1]$ belong to that plane? What is the span of p, q, r, s?

Exercise 6. Intersections of projective linear spaces ⋆ **CORE**

In $\mathbb{P}^3_{\Bbbk}$, consider the lines

$$L_1 = \mathbf{V}(x_2, x_3), \qquad L_2 = \mathbf{V}(x_0, x_1), \qquad L_3 = \mathbf{V}(x_1, x_3).$$

(i) Compute $L_1 \cap L_2$, $L_1 \cap L_3$, and $L_2 \cap L_3$.

(ii) Let

$$H_1 = \mathbf{V}(x_0 + x_1), \qquad H_2 = \mathbf{V}(x_2 - x_3).$$

Parametrise the line $H_1 \cap H_2$.

(iii) Explain, using vector-space dimensions, why two distinct projective lines in $\mathbb{P}^2_{\Bbbk}$ always meet, whereas two projective lines in $\mathbb{P}^3_{\Bbbk}$ may be disjoint.

Hint 1 (conceptual). A projective point is a non-zero vector up to scalar. A system of homogeneous linear equations has a projective solution precisely when its vector-space kernel is non-zero.

Exercise 7. Equations from determinants ⋆⋆ **CHOICE**

Let

$$p = [1:0:1], \qquad q = [0:1:1] \in \mathbb{P}^2_{\Bbbk}.$$

(i) Show that a point $x = [x_0 : x_1 : x_2]$ lies on the line through p and q exactly when

$$\det \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ x_0 & x_1 & x_2 \end{pmatrix} = 0.$$

(ii) Compute the resulting linear equation.

(iii) Formulate the analogous determinant criterion for the plane through three non-collinear points of $\mathbb{P}^3_{\Bbbk}$.

Exercise 8. Integral elements form a subring ⋆⋆ **CHOICE**

Let $A \to B$ be a homomorphism of commutative rings. Recall that $b \in B$ is *integral over* A if it satisfies a monic polynomial with coefficients in A.

(i) Let $\alpha \in B$ be integral over A, satisfying a monic equation of degree d. Show that $A[\alpha]$ is generated as an A-module by

$$1, \alpha, \ldots, \alpha^{d-1}.$$

(ii) Suppose that C is an A-algebra which is finitely generated as an A-module, say by $e_1, \ldots, e_r$, and let $c \in C$. Write

$$ce_i = \sum_j a_{ij} e_j, \qquad a_{ij} \in A.$$

Using the adjugate matrix, prove that c satisfies the monic equation

$$\det(TI_r - (a_{ij}))\big|_{T=c} = 0.$$

(iii) Let $\alpha, \beta \in B$ be integral over A. Explain why β is integral over $A[\alpha]$, and deduce that $A[\alpha, \beta]$ is a finite A-module.

(iv) Apply the preceding parts to prove that $\alpha + \beta$ and $\alpha\beta$ are integral over A. Show directly that $-\alpha$ is integral over A.

(v) Conclude that the elements of B integral over A form a subring of B containing the image of A.

Hint 1 (conceptual). The useful bridge is: an A-algebra which is finite as an A-module consists entirely of elements integral over A. For transitivity of finite generation, use generators u_i of $A[\alpha]$ over A and generators v_j of $A[\alpha, \beta]$ over $A[\alpha]$; then the products $u_i v_j$ generate over A.

Hint 2 (technical). If $E = (e_1, \ldots, e_r)^t$ and $M = (a_{ij})$, then $(cI_r - M)E = 0$. Multiply by $\mathrm{adj}(cI_r - M)$. The resulting determinant kills every generator e_i, hence kills $1 \in C$.

Exercise 9. Veronese and Segre coordinates ⋆⋆ **CHOICE**

(i) Show that

$$\nu_2 : \mathbb{P}^1_{\Bbbk} \longrightarrow \mathbb{P}^2_{\Bbbk}, \qquad [s : t] \longmapsto [s^2 : st : t^2]$$

is well defined and that its image is contained in the conic $x_0 x_2 - x_1^2 = 0$.

(ii) Prove that every point of this conic belongs to the image of ν_2.

(iii) Show that the Segre map

$$\sigma : \mathbb{P}^1_{\Bbbk} \times \mathbb{P}^1_{\Bbbk} \longrightarrow \mathbb{P}^3_{\Bbbk}, \quad ([s_0 : s_1], [t_0 : t_1]) \longmapsto [s_0 t_0 : s_0 t_1 : s_1 t_0 : s_1 t_1]$$

is well defined and that its image satisfies

$$z_{00} z_{11} - z_{01} z_{10} = 0.$$

(iv) Interpret the last equation as the condition that a 2×2 matrix has rank at most one, and deduce that every point of the quadric lies in the image of σ.

Hint 1 (conceptual). For the converse statements, use one non-zero coordinate to choose an affine chart. For Segre, arrange the four coordinates as a 2×2 matrix.

Hint 2 (technical). On the Veronese conic, if $x_0 \neq 0$, write the point as $[1 : x_1/x_0 : x_2/x_0]$ and use the equation to recover the last coordinate. Treat $x_0 = 0$ separately.